

1. Questions

Read the following information carefully and answer the questions based on it.

A student appeared for an examination in four different subjects namely Maths, Science, English and Hindi.

Maths: The total number of questions was $[50X]$, out of which $\frac{1}{5}^{\text{th}}$ were skipped. Out of the total number of questions attempted, the number of correct attempts was $[34X]$. The total marks obtained were 65.

Science: The total number of questions was 150, out of which $[10Y]\%$ were skipped. Out of the total number of questions attempted, the number of wrong attempts was 16. The total marks obtained were $[17Y + 2]$.

English: The total number of questions was $[M]$, out of which 25% is skipped. Out of the total number of questions attempted, the number of correct attempts was $[M - 28]$. The total marks obtained were $[9Y + 7X]$.

Hindi: The total number of questions was 120, out of which the number of questions skipped was $[N - 20]$. Out of the total number of questions attempted, the number of wrong attempts was $[M/4]$. The total marks obtained were $[2N - 5]$.

Note:

The total number of questions contained = the number of questions attempted + the number of questions skipped

The total number of questions attempted = the number of correct attempts + the number of wrong attempts

In each subject, 1 mark is awarded for each correct attempt and 0.25 mark is deducted for each wrong attempt.

If the ratio of the number of questions in the practical section of Maths and English is 2:1, and the number of questions in the theory section of Maths is 10 more than that in English, then find the number of questions in the practical section of English. (consider there are theory and practical section only)

- a. 20
- b. 10
- c. 30
- d. 40
- e. 50

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2. Questions

Find the average of the total number of correct attempts in all four subjects together.

- a. 52.25

- b. 44.75
- c. 80.20
- d. 30.80
- e. 68.50

3. Questions

If the total number of questions attempted by the student in computer subject is 45 and the marks obtained by him in the computer subject are 40. In computer subject, the number of wrong attempts is 11 less than the number of questions skipped, then find the difference between the total number of questions in computer and Maths.

- a. 32
- b. 66
- c. 40
- d. 24
- e. 50

4. Questions

In which of the following equation(s), both sides are equal?

Equation I: $10Y = N$

Equation II: $M = 40X$

Equation III: $N = 2M$

- a. Both I and II follow
- b. Only I follow
- c. Only III follow
- d. Both II and III follow
- e. All I, II, and III follow

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5. Questions

If the pass marks in Science are 15 more than that in Hindi and the ratio of the pass percentage in Science and Hindi is 6:5, then find the passing marks in Science.

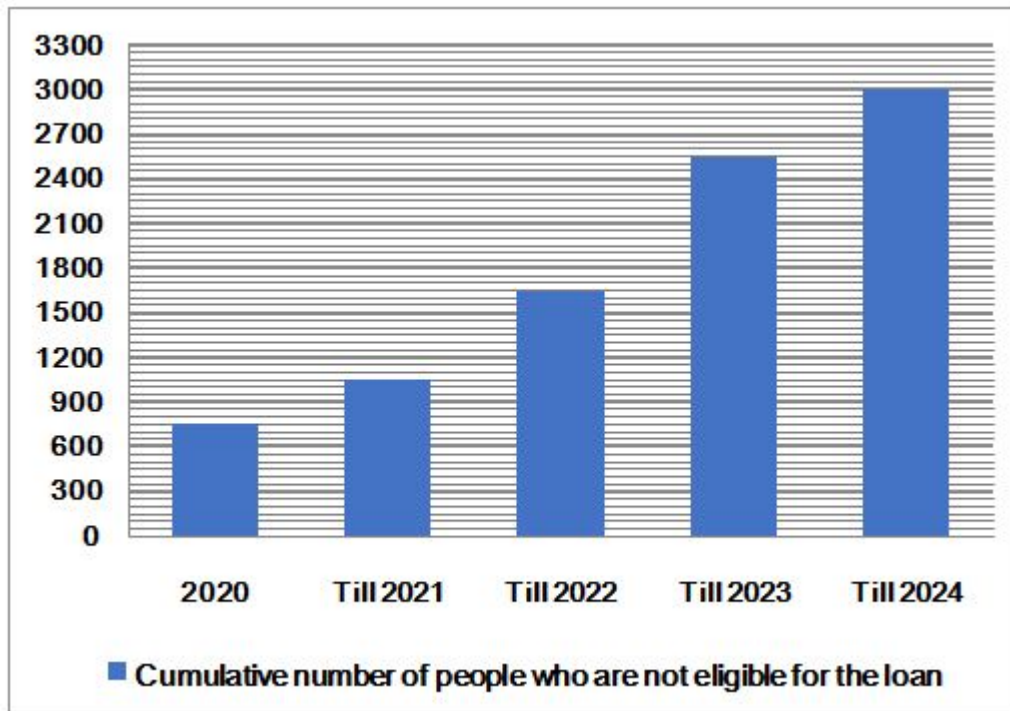
- a. 30
- b. 45
- c. 50
- d. 25

e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the cumulative number of people who are not eligible for a loan in a bank from 2020 till 2024.



Note:

The total number of people who enquired for the loan = the number of people who are eligible for the loan + the number of people who are not eligible for the loan

The total number of people who are eligible for the loan = the number of people who applied for the loan + the number of people who did not apply for the loan

The number of people who did not apply for the loan in 2020, 2021, 2022, 2023 and 2024 is 250, 200, 400, 800, and 300 respectively.

In 2020, the number of people who applied for the loan is $\frac{2}{3}$ rd of the total number of people who enquired for the loan.

The total number of people who enquired for the loan in 2021 is 20% of 2022.

The difference between the number of people who applied for the loan in 2021 and 2022 is 3500. (2021 < 2022)

The ratio of the total number of people who enquired for the loan in 2023 and 2024 is 10:7.

The number of people who applied for the loan in 2024 is 14.28% less than the previous year.

Find the number of people who applied for the loan in 2022 is approximately what percent of the total number of people who are eligible for the loan in the same year?

- a. 84%
- b. 60%
- c. 91%
- d. 78%
- e. 57%

7. Questions

In 2023, if $\frac{1}{3}^{\text{rd}}$ of the total number of people who enquired for the loan are females, while the number of males who are eligible for the loan is 2800, then find the ratio of the number of males and females who are not eligible for the loan in 2023.

- a. 1:4
- b. 2:7
- c. 3:8
- d. 5:9
- e. 1:1

8. Questions

If the number of people who did not apply for the loan in 2025 is $\frac{1}{5}^{\text{th}}$ of the total number of people who are eligible for the loan in the same year, while the number of people who applied for the loan in 2025 is 25% more than the previous year, then find the total number of people who are eligible for the loan in 2025.

- a. 3750
- b. 4550
- c. 5200
- d. 3350
- e. 6000

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9. Questions

Find the difference between the average number of people who enquired for the loan in 2021 and 2024 and the number of people who applied for the loan in 2020.

- a. 50
- b. 45
- c. 90
- d. 60

e. 75

10. Questions

In 2021, if 5% of the total number of people who applied for the loan are rejected and the number of people whose loan did not get disburse is 3 times of the number of people whose loan is rejected, then find the number of people whose loan get disbursed in 2021.

- a. 320
- b. 560
- c. 280
- d. 400
- e. 640

11. Questions

Article A is marked up by 50% above its cost price and sold at a discount of Rs. 400. Article B is sold at profit of Rs. 800 after allowing a discount of 14.28%. The ratio of the marked price of articles A and B is 6:7. The cost price of article A is Rs. 800 less than the selling price of article B.

Which of the following statement(s) is/are definitely true?

Statement I: The cost price of articles A and B is the same.

Statement II: Article A has been sold at a profit of 25%.

Statement III: Article B has been marked up by Rs. 1000.

- a. Both statements I and II
- b. Only statement II
- c. Only statement I
- d. Both statements II and III
- e. All statements I, II, and III

12. Questions

A car started from city X towards city Y but after 3 hours it stopped somewhere between city X and Y. Now, after few hours, the car started from the point where it was stopped and covered the remaining $\frac{1}{3}$ rd of the total distance with the reduced speed of 60 km/hr in 2 hours and finally reached city Y.

Which of the following statement(s) cannot be determined?

Statement I: The original speed of the car.

Statement II: The total time taken by the car to reach city Y.

Statement III: The total distance between cities X and Y.

- a. Both I and II
- b. Both II and III
- c. Only II
- d. Only I
- e. All I, II and III

13. Questions

The ratio of the expenditure and savings of P and Q is 2:1 and 5:1 respectively. The total income of P is Rs. 30000 more than that of Q, while if the savings of Q is increased by Rs. 15000, then the savings of P and Q will be equal.

Which of the following statement(s) is/are must be true?

Statement I: P spends 66.67% of his total income.

Statement II: The ratio of the savings of P and expenditure of Q is 4:5.

Statement III: The difference between the expenditure and savings of Q is Rs. 20000.

- a. Both statements I and II
- b. Only statement II
- c. Only statement I
- d. Both statements II and III
- e. All statements I, II and III

14. Questions

The amount invested in scheme A is Rs. 1200 at the rate of 20% per annum compounded annually, while the simple interest received from scheme B after 3 years is Rs. 78 less than the compound interest received from scheme A after 2 years.

Which of the following statement(s) can be determined?

I: The ratio of the amount invested in scheme A and B.

II: The rate of interest offered in scheme B.

III: The total interest received from scheme A and B together.

- a. Both II and III
- b. Only III
- c. Only I
- d. Both I and II
- e. All I, II and III

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15. Questions

The time taken by B and C together to complete the whole work is 12 days. C can complete the whole work in 30 days. A and B together can complete the same work in $(20/3)$ days.

Which of the following statement(s) is/are correct?

Statement I: A can complete the whole work alone in 10 days.

Statement II: B can complete 80% of the total work in 16 days.

Statement III: The time taken by A, B and C to complete the whole work is $(60/11)$ days.

- a. Both II and III
- b. Only III
- c. Only I
- d. Both I and II
- e. All I, II and III

16. Questions

The following question has two quantities as **Quantity I** and **Quantity II**. You have to determine the relationship between them and give an answer accordingly.

Quantity I: find the value of $[X/5]$. P started a business with the investment of Rs. $[X]$, after 8 months Q also joined him with the investment of Rs. $[X + 500]$. At the end of the year, the ratio of the profit share received by P and Q is 9:4.

Quantity II: find the value of $[Y^3]$. M started the business with Rs. 1600 and after $[Y]$ months, N also joined with 50% more investment than M. At the end of the year, the ratio of the profit share received by M and N is 8:5.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

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17. Questions

Quantity I: find the value of $[10n]$. Pipe A and C can alone can fill the tank in 12 minutes and 24 minutes respectively, whereas pipe B can empty the same tank alone in 30 minutes. Now, pipe A and B started to fill the tank together, but after $[2n + 3]$ minutes, both pipes were turned off and pipe C alone filled the remaining tank in $[n]$ minutes.

Quantity II: find the value of $[4m]$. The time taken by pipe X and Y to fill the tank alone is $[m]$ minutes and $[m + 5]$ minutes respectively. Now, pipe X and Y started to fill the tank together, but after 8 minutes,

pipe Y is turned off and pipe X alone filled the remaining tank in 1 minute. The total capacity of the tank is 60 ml. (m is positive integer)

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

18. Questions

Quantity I: Find the present age of the husband. The ratio of the present age of husband and wife is 19:16. The present age of their son is 3 years. At the time of their marriage, the ratio of the age of husband and wife was 11:9 and their son was born 2 years after their marriage.

Quantity II: The present age of P is what percent more than that of Q? The ratio of the present age of P and R is 3:2. The difference between the age of P and Q is same as the difference between the age of Q and R. P is elder than Q, who is elder than R.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

19. Questions

Quantity I: find the length of the platform. The time taken by train A of length 400 meters to cross the platform is same as the time taken by train B of length 300 meters to cross the same platform. The speed of train A and B is 120 m/sec and 100 m/sec respectively.

Quantity II: Find the four times of the speed of the train. If the time taken by a man to cross the train of length 225 meters running in opposite direction is 3 seconds. The ratio of the speed of the train and the man is 14:1.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

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20. Questions

Quantity I: Find the total population of the village. 25% of the total population of the village are illiterate,

which is equal to the number of post-graduates in the same village. The difference between the number of graduates and illiterates in the village is 500. (Only literates can be graduates and post-graduates)

Quantity II: Find the total number of employees in the company initially. The number of female employees in the company is 50% of the number of male employees in the same company. If 500 male employees left the company, which is equal to the number of newly joined female employees, then the total number of employees in the company will be 3000.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

Explanations:

1. Questions

Maths

The total number of questions = $50X$

The number of questions skipped = $1/5$ of $50X = 10X$

The number of questions attempted = $(50X - 10X) = 40X$

The number of correct attempts = $34X$

The number of wrong attempts = $(40X - 34X) = 6X$

The total marks obtained = $(1 * 34X) - (0.25 * 6X) = (34X - 1.50X) = 32.50X$

So,

$$32.50X = 65$$

$$X = 2$$

Therefore,

The total number of questions = $50X = (50 * 2) = 100$

The number of questions skipped = $10X = (10 * 2) = 20$

The number of questions attempted = $40X = (40 * 2) = 80$

The number of correct attempts = $34X = (34 * 2) = 68$

The number of wrong attempts = $6X = (6 * 2) = 12$

Science

The total number of questions = 150

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The number of questions skipped = $10Y\%$ of 150 = $15Y$

The number of questions attempted = $(150 - 15Y)$

The number of wrong attempts = 16

The number of correct attempts = $(150 - 15Y) - 16 = (134 - 15Y)$

The total marks obtained = $1 \times (134 - 15Y) - (0.25 \times 16) = (134 - 15Y) - 4 = (130 - 15Y)$

So,

$$130 - 15Y = 17Y + 2$$

$$32Y = 128$$

$$Y = 4$$

Therefore,

The total number of questions = 150

The number of questions skipped = $15Y = (15 \times 4) = 60$

The number of questions attempted = $(150 - 15Y) = (150 - 60) = 90$

The number of wrong attempts = 16

The number of correct attempts = $(134 - 15Y) = (134 - 60) = 74$

The total marks obtained = $(17Y + 2) = (17 \times 4 + 2) = (68 + 2) = 70$

English

The total number of questions = M

The number of questions skipped = 25% of M = $0.25M$

The number of questions attempted = $(M - 0.25M) = 0.75M$

The number of correct attempts = $(M - 28)$

The number of wrong attempts = $0.75M - (M - 28) = (28 - 0.25M)$

The total marks obtained = $1 \times (M - 28) - 0.25 \times (28 - 0.25M) = (M - 28) - (7 - 0.0625M) = (1.0625M - 35)$

So,

$$1.0625M - 35 = 9Y + 7X$$

$$1.0625M - 35 = (9 \times 4) + (7 \times 2)$$

$$1.0625M - 35 = 36 + 14$$

$$1.0625M = 50 + 35$$

$$1.0625M = 85$$

$$M = 80$$

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Therefore,

The total number of questions = $M = 80$

The number of questions skipped = $0.25M = (0.25 * 80) = 20$

The number of questions attempted = $0.75M = (0.75 * 80) = 60$

The number of correct attempts = $(M - 28) = (80 - 28) = 52$

The number of wrong attempts = $(28 - 0.25M) = (28 - 20) = 8$

The total marks obtained = $(9Y + 7X) = (9 * 4) + (7 * 2) = (36 + 14) = 50$

Hindi

The total number of questions = 120

The number of questions skipped = $(N - 20)$

The number of questions attempted = $120 - (N - 20) = (140 - N)$

The number of wrong attempts = $(M/4) = (80/4) = 20$

The number of correct attempts = $(140 - N) - 20 = (120 - N)$

The total marks obtained = $1 * (120 - N) - (0.25 * 20) = (120 - N) - 5 = (115 - N)$

So,

$$115 - N = 2N - 5$$

$$3N = 120$$

$$N = 40$$

Therefore,

The total number of questions = 120

The number of questions skipped = $(N - 20) = (40 - 20) = 20$

The number of questions attempted = $(140 - N) = (140 - 40) = 100$

The number of wrong attempts = 20

The number of correct attempts = $(120 - N) = (120 - 40) = 80$

The total marks obtained = $(2N - 5) = (2 * 40 - 5) = (80 - 5) = 75$

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Subject	The total number of questions	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts	The total marks obtained
Maths	100	80	20	68	12	65
Science	150	90	60	74	16	70
English	80	60	20	52	8	50
Hindi	120	100	20	80	20	75
Total	450	330	120	274	56	260

Answer: B

Now, according to the question,

The total number of questions contained in Maths = 100

The total number of questions contained in English = 80

Let the number of questions in the practical section in Maths and English be $2a$ and a respectively.

So, the number of questions contained in the theory section of Maths = $(100 - 2a)$

And, the number of questions contained in the theory section of English = $(80 - a)$

Now,

$$(100 - 2a) - (80 - a) = 10$$

$$20 - a = 10$$

$$a = 10$$

Therefore, the number of questions in the practical section of English = $a = 10$.

Hence, the correct answer is option B.

2. Questions

Maths

The total number of questions = $50X$

The number of questions skipped = $\frac{1}{5}$ of $50X = 10X$

The number of questions attempted = $(50X - 10X) = 40X$

The number of correct attempts = $34X$

The number of wrong attempts = $(40X - 34X) = 6X$

The total marks obtained = $(1 * 34X) - (0.25 * 6X) = (34X - 1.50X) = 32.50X$

So,

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$$32.50X = 65$$

$$X = 2$$

Therefore,

$$\text{The total number of questions} = 50X = (50 * 2) = 100$$

$$\text{The number of questions skipped} = 10X = (10 * 2) = 20$$

$$\text{The number of questions attempted} = 40X = (40 * 2) = 80$$

$$\text{The number of correct attempts} = 34X = (34 * 2) = 68$$

$$\text{The number of wrong attempts} = 6X = (6 * 2) = 12$$

Science

$$\text{The total number of questions} = 150$$

$$\text{The number of questions skipped} = 10Y\% \text{ of } 150 = 15Y$$

$$\text{The number of questions attempted} = (150 - 15Y)$$

$$\text{The number of wrong attempts} = 16$$

$$\text{The number of correct attempts} = (150 - 15Y) - 16 = (134 - 15Y)$$

$$\text{The total marks obtained} = 1 * (134 - 15Y) - (0.25 * 16) = (134 - 15Y) - 4 = (130 - 15Y)$$

So,

$$130 - 15Y = 17Y + 2$$

$$32Y = 128$$

$$Y = 4$$

Therefore,

$$\text{The total number of questions} = 150$$

$$\text{The number of questions skipped} = 15Y = (15 * 4) = 60$$

$$\text{The number of questions attempted} = (150 - 15Y) = (150 - 60) = 90$$

$$\text{The number of wrong attempts} = 16$$

$$\text{The number of correct attempts} = (134 - 15Y) = (134 - 60) = 74$$

$$\text{The total marks obtained} = (17Y + 2) = (17 * 4 + 2) = (68 + 2) = 70$$

English

$$\text{The total number of questions} = M$$

$$\text{The number of questions skipped} = 25\% \text{ of } M = 0.25M$$

$$\text{The number of questions attempted} = (M - 0.25M) = 0.75M$$

$$\text{The number of correct attempts} = (M - 28)$$

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The number of wrong attempts = $0.75M - (M - 28) = (28 - 0.25M)$

The total marks obtained = $1*(M - 28) - 0.25*(28 - 0.25M) = (M - 28) - (7 - 0.0625M) = (1.0625M - 35)$

So,

$$1.0625M - 35 = 9Y + 7X$$

$$1.0625M - 35 = (9 * 4) + (7 * 2)$$

$$1.0625M - 35 = 36 + 14$$

$$1.0625M = 50 + 35$$

$$1.0625M = 85$$

$$M = 80$$

Therefore,

The total number of questions = $M = 80$

The number of questions skipped = $0.25M = (0.25 * 80) = 20$

The number of questions attempted = $0.75M = (0.75 * 80) = 60$

The number of correct attempts = $(M - 28) = (80 - 28) = 52$

The number of wrong attempts = $(28 - 0.25M) = (28 - 20) = 8$

The total marks obtained = $(9Y + 7X) = (9 * 4) + (7 * 2) = (36 + 14) = 50$

Hindi

The total number of questions = 120

The number of questions skipped = $(N - 20)$

The number of questions attempted = $120 - (N - 20) = (140 - N)$

The number of wrong attempts = $(M/4) = (80/4) = 20$

The number of correct attempts = $(140 - N) - 20 = (120 - N)$

The total marks obtained = $1*(120 - N) - (0.25 * 20) = (120 - N) - 5 = (115 - N)$

So,

$$115 - N = 2N - 5$$

$$3N = 120$$

$$N = 40$$

Therefore,

The total number of questions = 120

The number of questions skipped = $(N - 20) = (40 - 20) = 20$

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The number of questions attempted = $(140 - N) = (140 - 40) = 100$

The number of wrong attempts = 20

The number of correct attempts = $(120 - N) = (120 - 40) = 80$

The total marks obtained = $(2N - 5) = (2 * 40 - 5) = (80 - 5) = 75$

Subject	The total number of questions	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts	The total marks obtained
Maths	100	80	20	68	12	65
Science	150	90	60	74	16	70
English	80	60	20	52	8	50
Hindi	120	100	20	80	20	75
Total	450	330	120	274	56	260

Answer: E

The number of correct attempts in,

Maths = 68

Science = 74

English = 52

Hindi = 80

So, the total number of correct attempts in all four subjects together = $(68 + 74 + 52 + 80) = 274$

Therefore, the required average = $(274/4) = 68.50$.

Hence, the correct answer is option E.

3. Questions

Maths

The total number of questions = $50X$

The number of questions skipped = $1/5$ of $50X = 10X$

The number of questions attempted = $(50X - 10X) = 40X$

The number of correct attempts = $34X$

The number of wrong attempts = $(40X - 34X) = 6X$

The total marks obtained = $(1 * 34X) - (0.25 * 6X) = (34X - 1.50X) = 32.50X$

So,

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$$32.50X = 65$$

$$X = 2$$

Therefore,

$$\text{The total number of questions} = 50X = (50 * 2) = 100$$

$$\text{The number of questions skipped} = 10X = (10 * 2) = 20$$

$$\text{The number of questions attempted} = 40X = (40 * 2) = 80$$

$$\text{The number of correct attempts} = 34X = (34 * 2) = 68$$

$$\text{The number of wrong attempts} = 6X = (6 * 2) = 12$$

Science

$$\text{The total number of questions} = 150$$

$$\text{The number of questions skipped} = 10Y\% \text{ of } 150 = 15Y$$

$$\text{The number of questions attempted} = (150 - 15Y)$$

$$\text{The number of wrong attempts} = 16$$

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So,

$$130 - 15Y = 17Y + 2$$

$$32Y = 128$$

$$Y = 4$$

Therefore,

$$\text{The total number of questions} = 150$$

$$\text{The number of questions skipped} = 15Y = (15 * 4) = 60$$

$$\text{The number of questions attempted} = (150 - 15Y) = (150 - 60) = 90$$

$$\text{The number of wrong attempts} = 16$$

$$\text{The number of correct attempts} = (134 - 15Y) = (134 - 60) = 74$$

$$\text{The total marks obtained} = (17Y + 2) = (17 * 4 + 2) = (68 + 2) = 70$$

English

$$\text{The total number of questions} = M$$

$$\text{The number of questions skipped} = 25\% \text{ of } M = 0.25M$$

$$\text{The number of questions attempted} = (M - 0.25M) = 0.75M$$

$$\text{The number of correct attempts} = (M - 28)$$

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$$\text{The number of wrong attempts} = 0.75M - (M - 28) = (28 - 0.25M)$$

$$\text{The total marks obtained} = 1*(M - 28) - 0.25*(28 - 0.25M) = (M - 28) - (7 - 0.0625M) = (1.0625M - 35)$$

So,

$$1.0625M - 35 = 9Y + 7X$$

$$1.0625M - 35 = (9 * 4) + (7 * 2)$$

$$1.0625M - 35 = 36 + 14$$

$$1.0625M = 50 + 35$$

$$1.0625M = 85$$

$$M = 80$$

Therefore,

$$\text{The total number of questions} = M = 80$$

$$\text{The number of questions skipped} = 0.25M = (0.25 * 80) = 20$$

$$\text{The number of questions attempted} = 0.75M = (0.75 * 80) = 60$$

$$\text{The number of correct attempts} = (M - 28) = (80 - 28) = 52$$

$$\text{The number of wrong attempts} = (28 - 0.25M) = (28 - 20) = 8$$

$$\text{The total marks obtained} = (9Y + 7X) = (9 * 4) + (7 * 2) = (36 + 14) = 50$$

Hindi

$$\text{The total number of questions} = 120$$

$$\text{The number of questions skipped} = (N - 20)$$

$$\text{The number of questions attempted} = 120 - (N - 20) = (140 - N)$$

$$\text{The number of wrong attempts} = (M/4) = (80/4) = 20$$

$$\text{The number of correct attempts} = (140 - N) - 20 = (120 - N)$$

$$\text{The total marks obtained} = 1*(120 - N) - (0.25 * 20) = (120 - N) - 5 = (115 - N)$$

So,

$$115 - N = 2N - 5$$

$$3N = 120$$

$$N = 40$$

Therefore,

$$\text{The total number of questions} = 120$$

$$\text{The number of questions skipped} = (N - 20) = (40 - 20) = 20$$

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The number of questions attempted = $(140 - N) = (140 - 40) = 100$

The number of wrong attempts = 20

The number of correct attempts = $(120 - N) = (120 - 40) = 80$

The total marks obtained = $(2N - 5) = (2 * 40 - 5) = (80 - 5) = 75$

Subject	The total number of questions	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts	The total marks obtained
Maths	100	80	20	68	12	65
Science	150	90	60	74	16	70
English	80	60	20	52	8	50
Hindi	120	100	20	80	20	75
Total	450	330	120	274	56	260

Answer: C

Computer

The total number of questions attempted = 45

Let the number of correct attempts be k .

So, the number of wrong attempts = $(45 - k)$

The total marks obtained = $(1 * k) - 0.25 * (45 - k) = k - (11.25 - 0.25k) = (1.25k - 11.25)$

Now,

$$1.25k - 11.25 = 40$$

$$1.25k = 51.25$$

$$k = 41$$

So, the number of wrong attempts = $(45 - k) = (45 - 41) = 4$

Now, number of questions skipped = $(4 + 11) = 15$

So, the total number of questions contained in the computer = $(45 + 15) = 60$.

And, the total number of questions in Maths = 100

Therefore, the required difference = $(100 - 60) = 40$.

Hence, the correct answer is option C.

4. Questions

Maths

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The total number of questions = $50X$

The number of questions skipped = $\frac{1}{5}$ of $50X = 10X$

The number of questions attempted = $(50X - 10X) = 40X$

The number of correct attempts = $34X$

The number of wrong attempts = $(40X - 34X) = 6X$

The total marks obtained = $(1 * 34X) - (0.25 * 6X) = (34X - 1.50X) = 32.50X$

So,

$$32.50X = 65$$

$$X = 2$$

Therefore,

The total number of questions = $50X = (50 * 2) = 100$

The number of questions skipped = $10X = (10 * 2) = 20$

The number of questions attempted = $40X = (40 * 2) = 80$

The number of correct attempts = $34X = (34 * 2) = 68$

The number of wrong attempts = $6X = (6 * 2) = 12$

Science

The total number of questions = 150

The number of questions skipped = $10Y\%$ of 150 = $15Y$

The number of questions attempted = $(150 - 15Y)$

The number of wrong attempts = 16

The number of correct attempts = $(150 - 15Y) - 16 = (134 - 15Y)$

The total marks obtained = $1 * (134 - 15Y) - (0.25 * 16) = (134 - 15Y) - 4 = (130 - 15Y)$

So,

$$130 - 15Y = 17Y + 2$$

$$32Y = 128$$

$$Y = 4$$

Therefore,

The total number of questions = 150

The number of questions skipped = $15Y = (15 * 4) = 60$

The number of questions attempted = $(150 - 15Y) = (150 - 60) = 90$

The number of wrong attempts = 16

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The number of correct attempts = $(134 - 15Y) = (134 - 60) = 74$

The total marks obtained = $(17Y + 2) = (17 * 4 + 2) = (68 + 2) = 70$

English

The total number of questions = M

The number of questions skipped = 25% of M = $0.25M$

The number of questions attempted = $(M - 0.25M) = 0.75M$

The number of correct attempts = $(M - 28)$

The number of wrong attempts = $0.75M - (M - 28) = (28 - 0.25M)$

The total marks obtained = $1*(M - 28) - 0.25*(28 - 0.25M) = (M - 28) - (7 - 0.0625M) = (1.0625M - 35)$

So,

$$1.0625M - 35 = 9Y + 7X$$

$$1.0625M - 35 = (9 * 4) + (7 * 2)$$

$$1.0625M - 35 = 36 + 14$$

$$1.0625M = 50 + 35$$

$$1.0625M = 85$$

$$M = 80$$

Therefore,

The total number of questions = M = 80

The number of questions skipped = $0.25M = (0.25 * 80) = 20$

The number of questions attempted = $0.75M = (0.75 * 80) = 60$

The number of correct attempts = $(M - 28) = (80 - 28) = 52$

The number of wrong attempts = $(28 - 0.25M) = (28 - 20) = 8$

The total marks obtained = $(9Y + 7X) = (9 * 4) + (7 * 2) = (36 + 14) = 50$

Hindi

@QuizByMickey

The total number of questions = 120

The number of questions skipped = $(N - 20)$

The number of questions attempted = $120 - (N - 20) = (140 - N)$

The number of wrong attempts = $(M/4) = (80/4) = 20$

The number of correct attempts = $(140 - N) - 20 = (120 - N)$

The total marks obtained = $1*(120 - N) - (0.25 * 20) = (120 - N) - 5 = (115 - N)$

So,

$$115 - N = 2N - 5$$

$$3N = 120$$

$$N = 40$$

Therefore,

The total number of questions = 120

The number of questions skipped = $(N - 20) = (40 - 20) = 20$

The number of questions attempted = $(140 - N) = (140 - 40) = 100$

The number of wrong attempts = 20

The number of correct attempts = $(120 - N) = (120 - 40) = 80$

The total marks obtained = $(2N - 5) = (2 * 40 - 5) = (80 - 5) = 75$

Subject	The total number of questions	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts	The total marks obtained
Maths	100	80	20	68	12	65
Science	150	90	60	74	16	70
English	80	60	20	52	8	50
Hindi	120	100	20	80	20	75
Total	450	330	120	274	56	260

Answer: A

From a common explanation, we got the values of,

$$X = 2$$

$$Y = 4$$

$$M = 80$$

$$N = 40$$

Equation I: $10Y = N$

$$10Y = (10 * 4) = 40$$

$$N = 40$$

So, both sides of equation I are equal.

Equation II: $M = 40X$

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$$M = 80$$

$$40X = (40 * 2) = 80$$

So, both sides of equation II are equal.

Equation III: $N = 2M$

$$N = 40$$

$$2M = (2 * 80) = 160$$

$$40 \neq 160$$

So, both sides of equation III are not equal.

Therefore, both I and II follow.

Hence, the correct answer is option A.

5. Questions

Maths

The total number of questions = $50X$

The number of questions skipped = $1/5$ of $50X = 10X$

The number of questions attempted = $(50X - 10X) = 40X$

The number of correct attempts = $34X$

The number of wrong attempts = $(40X - 34X) = 6X$

The total marks obtained = $(1 * 34X) - (0.25 * 6X) = (34X - 1.50X) = 32.50X$

So,

$$32.50X = 65$$

$$X = 2$$

Therefore,

The total number of questions = $50X = (50 * 2) = 100$

The number of questions skipped = $10X = (10 * 2) = 20$

The number of questions attempted = $40X = (40 * 2) = 80$

The number of correct attempts = $34X = (34 * 2) = 68$

The number of wrong attempts = $6X = (6 * 2) = 12$

Science

The total number of questions = 150

The number of questions skipped = $10Y\%$ of 150 = $15Y$

The number of questions attempted = $(150 - 15Y)$

@QuizByMickey

The number of wrong attempts = 16

The number of correct attempts = $(150 - 15Y) - 16 = (134 - 15Y)$

The total marks obtained = $1 \times (134 - 15Y) - (0.25 \times 16) = (134 - 15Y) - 4 = (130 - 15Y)$

So,

$$130 - 15Y = 17Y + 2$$

$$32Y = 128$$

$$Y = 4$$

Therefore,

The total number of questions = 150

The number of questions skipped = $15Y = (15 \times 4) = 60$

The number of questions attempted = $(150 - 15Y) = (150 - 60) = 90$

The number of wrong attempts = 16

The number of correct attempts = $(134 - 15Y) = (134 - 60) = 74$

The total marks obtained = $(17Y + 2) = (17 \times 4 + 2) = (68 + 2) = 70$

English

The total number of questions = M

The number of questions skipped = 25% of M = $0.25M$

The number of questions attempted = $(M - 0.25M) = 0.75M$

The number of correct attempts = $(M - 28)$

The number of wrong attempts = $0.75M - (M - 28) = (28 - 0.25M)$

The total marks obtained = $1 \times (M - 28) - 0.25 \times (28 - 0.25M) = (M - 28) - (7 - 0.0625M) = (1.0625M - 35)$

So,

$$1.0625M - 35 = 9Y + 7X$$

$$1.0625M - 35 = (9 \times 4) + (7 \times 2)$$

$$1.0625M - 35 = 36 + 14$$

$$1.0625M = 50 + 35$$

$$1.0625M = 85$$

$$M = 80$$

Therefore,

The total number of questions = M = 80

@QuizByMickey

The number of questions skipped = $0.25M = (0.25 * 80) = 20$

The number of questions attempted = $0.75M = (0.75 * 80) = 60$

The number of correct attempts = $(M - 28) = (80 - 28) = 52$

The number of wrong attempts = $(28 - 0.25M) = (28 - 20) = 8$

The total marks obtained = $(9Y + 7X) = (9 * 4) + (7 * 2) = (36 + 14) = 50$

Hindi

The total number of questions = 120

The number of questions skipped = $(N - 20)$

The number of questions attempted = $120 - (N - 20) = (140 - N)$

The number of wrong attempts = $(M/4) = (80/4) = 20$

The number of correct attempts = $(140 - N) - 20 = (120 - N)$

The total marks obtained = $1*(120 - N) - (0.25 * 20) = (120 - N) - 5 = (115 - N)$

So,

$$115 - N = 2N - 5$$

$$3N = 120$$

$$N = 40$$

Therefore,

The total number of questions = 120

The number of questions skipped = $(N - 20) = (40 - 20) = 20$

The number of questions attempted = $(140 - N) = (140 - 40) = 100$

The number of wrong attempts = 20

The number of correct attempts = $(120 - N) = (120 - 40) = 80$

The total marks obtained = $(2N - 5) = (2 * 40 - 5) = (80 - 5) = 75$

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Subject	The total number of questions	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts	The total marks obtained
Maths	100	80	20	68	12	65
Science	150	90	60	74	16	70
English	80	60	20	52	8	50
Hindi	120	100	20	80	20	75
Total	450	330	120	274	56	260

Answer: B

The maximum marks in science = $(150 * 1) = 150$

The maximum marks in Hindi = $(120 * 1) = 120$

Let the passing marks in Hindi be z .

So, the passing marks in Science = $(z + 15)$

Now, the passing percentage in,

Science = $(z + 15)/150 * 100 = (2z + 30)/3$

Hindi = $z/120 * 100 = 5z/6$

Now,

$(2z + 30)/3 / 5z/6 = 6/5$

$(10z + 150)/3 = 5z$

$10z + 150 = 15z$

$5z = 150$

$z = 30$

Therefore, the pass marks in science = $(z + 15) = (30 + 15) = 45$.

Hence, the correct answer is option B.

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6. Questions

The number of people who are not eligible for the loan in each year,

2020 = 750

2021 = $(1050 - 750) = 300$

2022 = $(1650 - 1050) = 600$

2023 = $(2550 - 1650) = 900$

$$2024 = (3000 - 2550) = 450$$

Now,

In 2020, the ratio of the total number of people who enquired for the loan and the number of people who applied for the loan = 3:2. Let it be $3a$ and $2a$ respectively.

Now,

The total number of people who enquired for the loan in 2020 = $3a$

The number of people who are not eligible for the loan in 2020 = 750

The number of people who are eligible for the loan in 2020 = $(3a - 750)$

The number of people who did not apply for the loan in 2020 = 250

The number of people who applied for the loan in 2020 = $(3a - 750) - 250 = (3a - 1000)$

So,

$$(3a - 1000) = 2a$$

$$a = 1000$$

2021 and 2022

The ratio of the total number of people who enquired for the loan in 2021 and 2022 = 20:100 = 1:5. Let it be b and $5b$ respectively.

Now,

The number of people who are not eligible for the loan in 2021 = 300

The number of people who are eligible for the loan in 2021 = $(b - 300)$

The number of people who did not apply for the loan in 2021 = 200

The number of people who applied for the loan in 2021 = $(b - 300) - 200 = (b - 500)$

Also,

The number of people who are not eligible for the loan in 2022 = 600

The number of people who are eligible for the loan in 2022 = $(5b - 600)$

The number of people who did not apply for the loan in 2022 = 400

The number of people who applied for the loan in 2022 = $(5b - 600) - 400 = (5b - 1000)$

So,

$$(5b - 1000) - (b - 500) = 3500$$

$$5b - 4500 = b - 500$$

$$4b = 4000$$

$$b = 1000$$

2023 and 2024

Let the total number of people who enquired for the loan in 2023 and 2024 be $10c$ and $7c$ respectively.

Now,

The number of people who are not eligible for the loan in 2023 = 900

The number of people who are eligible for the loan in 2023 = $(10c - 900)$

The number of people who did not apply for the loan in 2023 = 800

The number of people who applied for the loan in 2023 = $(10c - 900) - 800 = (10c - 1700)$

Also,

The number of people who are not eligible for the loan in 2024 = 450

The number of people who are eligible for the loan in 2024 = $(7c - 450)$

The number of people who did not apply for the loan in 2024 = 300

The number of people who applied for the loan in 2024 = $(7c - 450) - 300 = (7c - 750)$

Now, the ratio of the number of people who applied for the loan in 2023 and 2024 = 7:6. (Note: 14.28% = $1/7$)

So,

$$(10c - 1700) / (7c - 750) = 7/6$$

$$60c - 10200 = 49c - 5250$$

$$11c = 4950$$

$$c = 450$$

2020

The total number of people who enquired for the loan = $3a = (3 * 1000) = 3000$

The number of people who are not eligible for the loan = 750

The number of people who are eligible for the loan = $(3000 - 750) = 2250$

The number of people who did not apply for the loan = 250

The number of people who applied for the loan = $(2250 - 250) = 2000$

2021

The total number of people who enquired for the loan = $b = 1000$

The number of people who are not eligible for the loan = 300

The number of people who are eligible for the loan = $(1000 - 300) = 700$

The number of people who did not apply for the loan = 200

The number of people who applied for the loan = $(700 - 200) = 500$

2022

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The total number of people who enquired for the loan = $5b = (5 * 1000) = 5000$

The number of people who are not eligible for the loan = 600

The number of people who are eligible for the loan = $(5000 - 600) = 4400$

The number of people who did not apply for the loan = 400

The number of people who applied for the loan = $(4400 - 400) = 4000$

2023

The total number of people who enquired for the loan = $10c = (10 * 450) = 4500$

The number of people who are not eligible for the loan = 900

The number of people who are eligible for the loan = $(4500 - 900) = 3600$

The number of people who did not apply for the loan = 800

The number of people who applied for the loan = $(3600 - 800) = 2800$

2024

The total number of people who enquired for the loan = $7c = (7 * 450) = 3150$

The number of people who are not eligible for the loan = 450

The number of people who are eligible for the loan = $(3150 - 450) = 2700$

The number of people who did not apply for the loan = 300

The number of people who applied for the loan = $(2700 - 300) = 2400$

Year	The total number of people who enquired	The total number of people who are eligible	The total number of people who are not eligible	The total number of people who applied	The total number of people who did not apply
2020	3000	2250	750	2000	250
2021	1000	700	300	500	200
2022	5000	4400	600	4000	400
2023	4500	3600	900	2800	800
2024	3150	2700	450	2400	300
Total	16650	13650	3000	11700	1950

Answer: C

Now, according to the question,

2022

The number of people who applied for the loan = 4000

The total number of people who are eligible for the loan = 4400.

Therefore, the required percentage = $4000/4400 \times 100 = 90.90 \approx 91\%$.

Hence, the correct answer is option C.

7. Questions

The number of people who are not eligible for the loan in each year,

$$2020 = 750$$

$$2021 = (1050 - 750) = 300$$

$$2022 = (1650 - 1050) = 600$$

$$2023 = (2550 - 1650) = 900$$

$$2024 = (3000 - 2550) = 450$$

Now,

In 2020, the ratio of the total number of people who enquired for the loan and the number of people who applied for the loan = 3:2. Let it be $3a$ and $2a$ respectively.

Now,

The total number of people who enquired for the loan in 2020 = $3a$

The number of people who are not eligible for the loan in 2020 = 750

The number of people who are eligible for the loan in 2020 = $(3a - 750)$

The number of people who did not apply for the loan in 2020 = 250

The number of people who applied for the loan in 2020 = $(3a - 750) - 250 = (3a - 1000)$

So,

$$(3a - 1000) = 2a$$

$$a = 1000$$

2021 and 2022

The ratio of the total number of people who enquired for the loan in 2021 and 2022 = 20:100 = 1:5. Let it be b and $5b$ respectively.

Now,

The number of people who are not eligible for the loan in 2021 = 300

The number of people who are eligible for the loan in 2021 = $(b - 300)$

The number of people who did not apply for the loan in 2021 = 200

The number of people who applied for the loan in 2021 = $(b - 300) - 200 = (b - 500)$

Also,

The number of people who are not eligible for the loan in 2022 = 600

The number of people who are eligible for the loan in 2022 = $(5b - 600)$

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The number of people who did not apply for the loan in 2022 = 400

The number of people who applied for the loan in 2022 = $(5b - 600) - 400 = (5b - 1000)$

So,

$$(5b - 1000) - (b - 500) = 3500$$

$$5b - 4500 = b - 500$$

$$4b = 4000$$

$$b = 1000$$

2023 and 2024

Let the total number of people who enquired for the loan in 2023 and 2024 be $10c$ and $7c$ respectively.

Now,

The number of people who are not eligible for the loan in 2023 = 900

The number of people who are eligible for the loan in 2023 = $(10c - 900)$

The number of people who did not apply for the loan in 2023 = 800

The number of people who applied for the loan in 2023 = $(10c - 900) - 800 = (10c - 1700)$

Also,

The number of people who are not eligible for the loan in 2024 = 450

The number of people who are eligible for the loan in 2024 = $(7c - 450)$

The number of people who did not apply for the loan in 2024 = 300

The number of people who applied for the loan in 2024 = $(7c - 450) - 300 = (7c - 750)$

Now, the ratio of the number of people who applied for the loan in 2023 and 2024 = 7:6. (Note: 14.28% = 1/7)

So,

$$(10c - 1700) / (7c - 750) = 7/6$$

$$60c - 10200 = 49c - 5250$$

$$11c = 4950$$

$$c = 450$$

2020

The total number of people who enquired for the loan = $3a = (3 * 1000) = 3000$

The number of people who are not eligible for the loan = 750

The number of people who are eligible for the loan = $(3000 - 750) = 2250$

The number of people who did not apply for the loan = 250

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The number of people who applied for the loan = $(2250 - 250) = 2000$

2021

The total number of people who enquired for the loan = $b = 1000$

The number of people who are not eligible for the loan = 300

The number of people who are eligible for the loan = $(1000 - 300) = 700$

The number of people who did not apply for the loan = 200

The number of people who applied for the loan = $(700 - 200) = 500$

2022

The total number of people who enquired for the loan = $5b = (5 * 1000) = 5000$

The number of people who are not eligible for the loan = 600

The number of people who are eligible for the loan = $(5000 - 600) = 4400$

The number of people who did not apply for the loan = 400

The number of people who applied for the loan = $(4400 - 400) = 4000$

2023

The total number of people who enquired for the loan = $10c = (10 * 450) = 4500$

The number of people who are not eligible for the loan = 900

The number of people who are eligible for the loan = $(4500 - 900) = 3600$

The number of people who did not apply for the loan = 800

The number of people who applied for the loan = $(3600 - 800) = 2800$

2024

The total number of people who enquired for the loan = $7c = (7 * 450) = 3150$

The number of people who are not eligible for the loan = 450

The number of people who are eligible for the loan = $(3150 - 450) = 2700$

The number of people who did not apply for the loan = 300

The number of people who applied for the loan = $(2700 - 300) = 2400$

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Year	The total number of people who enquired	The total number of people who are eligible	The total number of people who are not eligible	The total number of people who applied	The total number of people who did not apply
2020	3000	2250	750	2000	250
2021	1000	700	300	500	200
2022	5000	4400	600	4000	400
2023	4500	3600	900	2800	800
2024	3150	2700	450	2400	300
Total	16650	13650	3000	11700	1950

Answer: B

2023

The total number of people who are enquired for the loan = 4500

The number of females who enquired for the loan = $\frac{1}{3}$ of 450 = 1500

The number of males who enquired for the loan = $(4500 - 1500) = 3000$

The total number of people who are eligible for the loan = 3600

The number of males who are eligible for the loan = 2800

The number of females who are eligible for the loan = $(3600 - 2800) = 800$

The number of males who are not eligible for the loan = $(3000 - 2800) = 200$

The number of females who are not eligible for the loan = $(1500 - 800) = 700$

Therefore, the required ratio = $200:700 = 2:7$.

Hence, the correct answer is option B.

8. Questions

The number of people who are not eligible for the loan in each year,

2020 = 750

2021 = $(1050 - 750) = 300$

2022 = $(1650 - 1050) = 600$

2023 = $(2550 - 1650) = 900$

2024 = $(3000 - 2550) = 450$

Now,

In 2020, the ratio of the total number of people who enquired for the loan and the number of people who applied for the loan = 3:2. Let it be 3a and 2a respectively.

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Now,

The total number of people who enquired for the loan in 2020 = $3a$

The number of people who are not eligible for the loan in 2020 = 750

The number of people who are eligible for the loan in 2020 = $(3a - 750)$

The number of people who did not apply for the loan in 2020 = 250

The number of people who applied for the loan in 2020 = $(3a - 750) - 250 = (3a - 1000)$

So,

$$(3a - 1000) = 2a$$

$$a = 1000$$

2021 and 2022

The ratio of the total number of people who enquired for the loan in 2021 and 2022 = $20:100 = 1:5$. Let it be b and $5b$ respectively.

Now,

The number of people who are not eligible for the loan in 2021 = 300

The number of people who are eligible for the loan in 2021 = $(b - 300)$

The number of people who did not apply for the loan in 2021 = 200

The number of people who applied for the loan in 2021 = $(b - 300) - 200 = (b - 500)$

Also,

The number of people who are not eligible for the loan in 2022 = 600

The number of people who are eligible for the loan in 2022 = $(5b - 600)$

The number of people who did not apply for the loan in 2022 = 400

The number of people who applied for the loan in 2022 = $(5b - 600) - 400 = (5b - 1000)$

So,

$$(5b - 1000) - (b - 500) = 3500$$

$$5b - 4500 = b - 500$$

$$4b = 4000$$

$$b = 1000$$

2023 and 2024

Let the total number of people who enquired for the loan in 2023 and 2024 be $10c$ and $7c$ respectively.

Now,

The number of people who are not eligible for the loan in 2023 = 900

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The number of people who are eligible for the loan in 2023 = $(10c - 900)$

The number of people who did not apply for the loan in 2023 = 800

The number of people who applied for the loan in 2023 = $(10c - 900) - 800 = (10c - 1700)$

Also,

The number of people who are not eligible for the loan in 2024 = 450

The number of people who are eligible for the loan in 2024 = $(7c - 450)$

The number of people who did not apply for the loan in 2024 = 300

The number of people who applied for the loan in 2024 = $(7c - 450) - 300 = (7c - 750)$

Now, the ratio of the number of people who applied for the loan in 2023 and 2024 = 7:6. (Note: 14.28% = 1/7)

So,

$$(10c - 1700) / (7c - 750) = 7/6$$

$$60c - 10200 = 49c - 5250$$

$$11c = 4950$$

$$c = 450$$

2020

The total number of people who enquired for the loan = $3a = (3 * 1000) = 3000$

The number of people who are not eligible for the loan = 750

The number of people who are eligible for the loan = $(3000 - 750) = 2250$

The number of people who did not apply for the loan = 250

The number of people who applied for the loan = $(2250 - 250) = 2000$

2021

The total number of people who enquired for the loan = $b = 1000$

The number of people who are not eligible for the loan = 300

The number of people who are eligible for the loan = $(1000 - 300) = 700$

The number of people who did not apply for the loan = 200

The number of people who applied for the loan = $(700 - 200) = 500$

2022

The total number of people who enquired for the loan = $5b = (5 * 1000) = 5000$

The number of people who are not eligible for the loan = 600

The number of people who are eligible for the loan = $(5000 - 600) = 4400$

The number of people who did not apply for the loan = 400

The number of people who applied for the loan = $(4400 - 400) = 4000$

2023

The total number of people who enquired for the loan = $10c = (10 * 450) = 4500$

The number of people who are not eligible for the loan = 900

The number of people who are eligible for the loan = $(4500 - 900) = 3600$

The number of people who did not apply for the loan = 800

The number of people who applied for the loan = $(3600 - 800) = 2800$

2024

The total number of people who enquired for the loan = $7c = (7 * 450) = 3150$

The number of people who are not eligible for the loan = 450

The number of people who are eligible for the loan = $(3150 - 450) = 2700$

The number of people who did not apply for the loan = 300

The number of people who applied for the loan = $(2700 - 300) = 2400$

Year	The total number of people who enquired	The total number of people who are eligible	The total number of people who are not eligible	The total number of people who applied	The total number of people who did not apply
2020	3000	2250	750	2000	250
2021	1000	700	300	500	200
2022	5000	4400	600	4000	400
2023	4500	3600	900	2800	800
2024	3150	2700	450	2400	300
Total	16650	13650	3000	11700	1950

Answer: A

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2025

The number of people who applied for the loan = $125\% \text{ of } 2400 = 3000$

The ratio of the number of people who applied and those who did not apply for the loan = 4:1.

The number of people who did not apply for the loan = $3000/4*1 = 750$

Therefore, the total number of people who are eligible for the loan in 2025 = $(3000 + 750) = 3750$.

Hence, the correct answer is option A.

9. Questions

The number of people who are not eligible for the loan in each year,

$$2020 = 750$$

$$2021 = (1050 - 750) = 300$$

$$2022 = (1650 - 1050) = 600$$

$$2023 = (2550 - 1650) = 900$$

$$2024 = (3000 - 2550) = 450$$

Now,

In 2020, the ratio of the total number of people who enquired for the loan and the number of people who applied for the loan = 3:2. Let it be $3a$ and $2a$ respectively.

Now,

The total number of people who enquired for the loan in 2020 = $3a$

The number of people who are not eligible for the loan in 2020 = 750

The number of people who are eligible for the loan in 2020 = $(3a - 750)$

The number of people who did not apply for the loan in 2020 = 250

The number of people who applied for the loan in 2020 = $(3a - 750) - 250 = (3a - 1000)$

So,

$$(3a - 1000) = 2a$$

$$a = 1000$$

2021 and 2022

The ratio of the total number of people who enquired for the loan in 2021 and 2022 = 20:100 = 1:5. Let it be b and $5b$ respectively.

Now,

The number of people who are not eligible for the loan in 2021 = 300

The number of people who are eligible for the loan in 2021 = $(b - 300)$

The number of people who did not apply for the loan in 2021 = 200

The number of people who applied for the loan in 2021 = $(b - 300) - 200 = (b - 500)$

Also,

The number of people who are not eligible for the loan in 2022 = 600

The number of people who are eligible for the loan in 2022 = $(5b - 600)$

The number of people who did not apply for the loan in 2022 = 400

The number of people who applied for the loan in 2022 = $(5b - 600) - 400 = (5b - 1000)$

So,

$$(5b - 1000) - (b - 500) = 3500$$

$$5b - 4500 = b - 500$$

$$4b = 4000$$

$$b = 1000$$

2023 and 2024

Let the total number of people who enquired for the loan in 2023 and 2024 be $10c$ and $7c$ respectively.

Now,

The number of people who are not eligible for the loan in 2023 = 900

The number of people who are eligible for the loan in 2023 = $(10c - 900)$

The number of people who did not apply for the loan in 2023 = 800

The number of people who applied for the loan in 2023 = $(10c - 900) - 800 = (10c - 1700)$

Also,

The number of people who are not eligible for the loan in 2024 = 450

The number of people who are eligible for the loan in 2024 = $(7c - 450)$

The number of people who did not apply for the loan in 2024 = 300

The number of people who applied for the loan in 2024 = $(7c - 450) - 300 = (7c - 750)$

Now, the ratio of the number of people who applied for the loan in 2023 and 2024 = 7:6. (Note: 14.28% = $1/7$)

So,

$$(10c - 1700) / (7c - 750) = 7/6$$

$$60c - 10200 = 49c - 5250$$

$$11c = 4950$$

$$c = 450$$

2020

The total number of people who enquired for the loan = $3a = (3 * 1000) = 3000$

The number of people who are not eligible for the loan = 750

The number of people who are eligible for the loan = $(3000 - 750) = 2250$

The number of people who did not apply for the loan = 250

The number of people who applied for the loan = $(2250 - 250) = 2000$

2021

The total number of people who enquired for the loan = $b = 1000$

The number of people who are not eligible for the loan = 300

The number of people who are eligible for the loan = $(1000 - 300) = 700$

The number of people who did not apply for the loan = 200

The number of people who applied for the loan = $(700 - 200) = 500$

2022

The total number of people who enquired for the loan = $5b = (5 * 1000) = 5000$

The number of people who are not eligible for the loan = 600

The number of people who are eligible for the loan = $(5000 - 600) = 4400$

The number of people who did not apply for the loan = 400

The number of people who applied for the loan = $(4400 - 400) = 4000$

2023

The total number of people who enquired for the loan = $10c = (10 * 450) = 4500$

The number of people who are not eligible for the loan = 900

The number of people who are eligible for the loan = $(4500 - 900) = 3600$

The number of people who did not apply for the loan = 800

The number of people who applied for the loan = $(3600 - 800) = 2800$

2024

The total number of people who enquired for the loan = $7c = (7 * 450) = 3150$

The number of people who are not eligible for the loan = 450

The number of people who are eligible for the loan = $(3150 - 450) = 2700$

The number of people who did not apply for the loan = 300

The number of people who applied for the loan = $(2700 - 300) = 2400$

Year	The total number of people who enquired	The total number of people who are eligible	The total number of people who are not eligible	The total number of people who applied	The total number of people who did not apply
2020	3000	2250	750	2000	250
2021	1000	700	300	500	200
2022	5000	4400	600	4000	400
2023	4500	3600	900	2800	800
2024	3150	2700	450	2400	300
Total	16650	13650	3000	11700	1950

Answer: E

The total number of people who enquired for the loan in 2021 = 1000

The total number of people who enquired for the loan in 2024 = 3150

So, the average number of people who enquired for the loan in 2021 and 2024 = $(1000 + 3150)/2 = 4150/2 = 2075$

And, the number of people who applied for the loan in 2020 = 2000

Therefore, the required difference = $(2075 - 2000) = 75$.

Hence, the correct answer is option E.

10. Questions

The number of people who are not eligible for the loan in each year,

2020 = 750

2021 = $(1050 - 750) = 300$

2022 = $(1650 - 1050) = 600$

2023 = $(2550 - 1650) = 900$

2024 = $(3000 - 2550) = 450$

Now,

In 2020, the ratio of the total number of people who enquired for the loan and the number of people who applied for the loan = 3:2. Let it be $3a$ and $2a$ respectively.

Now,

The total number of people who enquired for the loan in 2020 = $3a$

The number of people who are not eligible for the loan in 2020 = 750

The number of people who are eligible for the loan in 2020 = $(3a - 750)$

The number of people who did not apply for the loan in 2020 = 250

The number of people who applied for the loan in 2020 = $(3a - 750) - 250 = (3a - 1000)$

So,

$(3a - 1000) = 2a$

$a = 1000$

2021 and 2022

The ratio of the total number of people who enquired for the loan in 2021 and 2022 = 20:100 = 1:5. Let it be b and $5b$ respectively.

Now,

The number of people who are not eligible for the loan in 2021 = 300

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The number of people who are eligible for the loan in 2021 = $(b - 300)$

The number of people who did not apply for the loan in 2021 = 200

The number of people who applied for the loan in 2021 = $(b - 300) - 200 = (b - 500)$

Also,

The number of people who are not eligible for the loan in 2022 = 600

The number of people who are eligible for the loan in 2022 = $(5b - 600)$

The number of people who did not apply for the loan in 2022 = 400

The number of people who applied for the loan in 2022 = $(5b - 600) - 400 = (5b - 1000)$

So,

$$(5b - 1000) - (b - 500) = 3500$$

$$5b - 4500 = b - 500$$

$$4b = 4000$$

$$b = 1000$$

2023 and 2024

Let the total number of people who enquired for the loan in 2023 and 2024 be $10c$ and $7c$ respectively.

Now,

The number of people who are not eligible for the loan in 2023 = 900

The number of people who are eligible for the loan in 2023 = $(10c - 900)$

The number of people who did not apply for the loan in 2023 = 800

The number of people who applied for the loan in 2023 = $(10c - 900) - 800 = (10c - 1700)$

Also,

The number of people who are not eligible for the loan in 2024 = 450

The number of people who are eligible for the loan in 2024 = $(7c - 450)$

The number of people who did not apply for the loan in 2024 = 300

The number of people who applied for the loan in 2024 = $(7c - 450) - 300 = (7c - 750)$

Now, the ratio of the number of people who applied for the loan in 2023 and 2024 = 7:6. (Note: 14.28% = 1/7)

So,

$$(10c - 1700) / (7c - 750) = 7/6$$

$$60c - 10200 = 49c - 5250$$

$$11c = 4950$$

$$c = 450$$

2020

The total number of people who enquired for the loan = $3a = (3 * 1000) = 3000$

The number of people who are not eligible for the loan = 750

The number of people who are eligible for the loan = $(3000 - 750) = 2250$

The number of people who did not apply for the loan = 250

The number of people who applied for the loan = $(2250 - 250) = 2000$

2021

The total number of people who enquired for the loan = $b = 1000$

The number of people who are not eligible for the loan = 300

The number of people who are eligible for the loan = $(1000 - 300) = 700$

The number of people who did not apply for the loan = 200

The number of people who applied for the loan = $(700 - 200) = 500$

2022

The total number of people who enquired for the loan = $5b = (5 * 1000) = 5000$

The number of people who are not eligible for the loan = 600

The number of people who are eligible for the loan = $(5000 - 600) = 4400$

The number of people who did not apply for the loan = 400

The number of people who applied for the loan = $(4400 - 400) = 4000$

2023

The total number of people who enquired for the loan = $10c = (10 * 450) = 4500$

The number of people who are not eligible for the loan = 900

The number of people who are eligible for the loan = $(4500 - 900) = 3600$

The number of people who did not apply for the loan = 800

The number of people who applied for the loan = $(3600 - 800) = 2800$

2024

The total number of people who enquired for the loan = $7c = (7 * 450) = 3150$

The number of people who are not eligible for the loan = 450

The number of people who are eligible for the loan = $(3150 - 450) = 2700$

The number of people who did not apply for the loan = 300

The number of people who applied for the loan = $(2700 - 300) = 2400$

Year	The total number of people who enquired	The total number of people who are eligible	The total number of people who are not eligible	The total number of people who applied	The total number of people who did not apply
2020	3000	2250	750	2000	250
2021	1000	700	300	500	200
2022	5000	4400	600	4000	400
2023	4500	3600	900	2800	800
2024	3150	2700	450	2400	300
Total	16650	13650	3000	11700	1950

Answer: D

2021

The total number of people who applied for the loan = 500

The number of people whose loan is rejected = 5% of 500 = 25

So, the total number of people whose loan is accepted = $(500 - 25) = 475$

The number of people whose loan did not get disburse = $(25 * 3) = 75$

Therefore, the number of people whose loan get disbursed = $(475 - 75) = 400$.

Hence, the correct answer is option D.

11. Questions

Answer: A

Let the marked price of articles A and B be Rs. $6n$ and Rs. $7n$ respectively.

Article A

The M.P. = Rs. $6n$

The ratio of M.P and C.P. = $150:100 = 3:2$.

The C.P. = $6n/3*2 = \text{Rs. } 4n$

The S.P. = Rs. $(6n - 400)$

Article B

The M.P. = $6n/6*7 = \text{Rs. } 7n$

The S.P. = $6/7$ of $7n = \text{Rs. } 6n$ (**Note: $14.28\% = 1/7$**)

The C.P. = Rs. $(6n - 800)$

Now,

$$6n - 4n = 800$$

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$$2n = 800$$

$$n = 400$$

Article A

$$\text{The M.P.} = 6n = (6 * 400) = \text{Rs. } 2400$$

$$\text{The C.P.} = 4n = (4 * 400) = \text{Rs. } 1600$$

$$\text{The S.P.} = (6n - 400) = (2400 - 400) = \text{Rs. } 2000$$

Article B

$$\text{The M.P.} = 7n = (7 * 400) = \text{Rs. } 2800$$

$$\text{The S.P.} = 6n = \text{Rs. } 2400$$

$$\text{The C.P.} = (6n - 800) = (2400 - 800) = \text{Rs. } 1600$$

Statement I

$$\text{The C.P. of article A} = \text{Rs. } 1600$$

$$\text{The C.P. of article B} = \text{Rs. } 1600$$

So, statement I is definitely true.

Statement II

$$\text{The C.P. of article A} = \text{Rs. } 1600$$

$$\text{The S.P. of article A} = \text{Rs. } 2000$$

$$\text{The profit earned on selling article A} = (2000 - 1600) = \text{Rs. } 400$$

$$\text{The profit percentage on selling article A} = (400/1600 * 100)\% = 25\%.$$

So, statement II is definitely true.

Statement III

$$\text{The M.P. of article B} = \text{Rs. } 2800$$

$$\text{The C.P. of article B} = \text{Rs. } 1600$$

$$\text{The mark-up amount on article B} = (2800 - 1600) = \text{Rs. } 1200$$

So, statement III is definitely not true.

Therefore, statements I and II are definitely true.

Hence, the correct answer is option A.

Time, Speed and Distance

12. Questions

Answer: C

The ratio of the distance covered by the car in the first and second parts = 2:1

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The distance covered by the car in second part of the journey = $(60 * 2) = 120$ km.

So, the distance covered by the car in the first part of the journey = $(120/1)*2 = 240$ km

Therefore, the total distance covered = $(120 + 240) = 360$ km

Now, let the original speed of the car to cover the first part of the journey be V km/hr

The time taken to cover the distance during the first part of the journey = 3 hours

So,

$$(V * 3) = 240$$

$$3V = 240$$

$$V = 80$$

Therefore, the original speed of the car = $V = 80$ km/hr

I: The original speed of the car = 80 km/hr

Statement I can be determined.

II: The total time taken by the car to cover the total distance between city X and Y cannot be determined, as we do not have the time duration that the car has spent when it was stopped somewhere between city X and Y.

Statement II cannot be determined.

III: The total distance between cities X and Y = 360 km

Statement III can be determined.

Therefore, only statement II cannot be determined.

Hence, the correct answer is option C.

13. Questions

Answer: E

Let the expenditure and savings of P be Rs. $2x$ and Rs. x respectively.

Also, let the expenditure and savings of Q be Rs. $5y$ and Rs. y respectively.

So, the total income of P = $(2x + x) = \text{Rs. } 3x$

And, the total income of Q = $(5y + y) = \text{Rs. } 6y$

Now,

$$3x - 6y = 30000$$

$$x - 2y = 10000 \dots\dots\dots(i)$$

And,

$$y + 15000 = x$$

$$x - y = 15000 \dots\dots\dots(ii)$$

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By solving equation (i) and (ii), we get,

$$y = 5000$$

Now, put the value of y in equation (i), we get,

$$x - 2y = 10000$$

$$x - (2 * 5000) = 10000$$

$$x = 10000 + 10000$$

$$x = 20000$$

For P

$$\text{The total income} = 3x = (3 * 20000) = \text{Rs. } 60000$$

$$\text{The expenditure} = 2x = (2 * 20000) = \text{Rs. } 40000$$

$$\text{The savings} = x = \text{Rs. } 20000$$

For Q

$$\text{The total income} = 6y = (6 * 5000) = \text{Rs. } 30000$$

$$\text{The expenditure} = 5y = (5 * 5000) = \text{Rs. } 25000$$

$$\text{The savings} = y = \text{Rs. } 5000$$

Statement I

$$\text{The expenditure of P} = \text{Rs. } 40000$$

$$\text{The total income of P} = \text{Rs. } 60000$$

$$\text{The required percentage} = (40000/60000 * 100)\% = 66.67\%$$

So, statement I is true.

Statement II

$$\text{The savings of P} = \text{Rs. } 20000$$

$$\text{The expenditure of Q} = \text{Rs. } 25000$$

$$\text{The required ratio} = 4:5$$

So, statement II is true.

Statement III

$$\text{The expenditure of Q} = \text{Rs. } 25000$$

$$\text{The savings of Q} = \text{Rs. } 5000$$

So, statement III is true.

Therefore, all statements I, II and III must be true.

Hence, the correct answer is option E.

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14. Questions

Answer: B

The amount invested in scheme A = Rs. 1200

The rate of interest in scheme A (C.I.) = $(20 + 20) + (20 * 20)/100 = (40 + 4)\% = 44\%$

So, the compound interest received from scheme A = 44% of 1200 = 528

Therefore, the simple interest received from scheme B = $(528 - 78) = \text{Rs. } 450$

Here, the rate of interest is not given in the question, so we cannot determine the amount invested in scheme B.

Statement I

The ratio of the amount invested in scheme A and B cannot be determined, as we do not have the amount invested in scheme B.

So, statement I cannot be determined.

Statement II

The rate of interest offered in scheme B cannot be determined.

So, statement II cannot be determined.

Statement III

The total interest received from scheme A and B together = $(528 + 450) = \text{Rs. } 978$.

So, statement III can be determined.

Therefore, only III can be determined.

Hence, the correct answer is option B.

15. Questions

Answer: E

The time taken to complete the whole work by,

$$(B + C) = 12 \text{ days}$$

$$C = 30 \text{ days}$$

$$(A + B) = (20/3) \text{ days}$$

The total work = 60 units
(L.C.M. of the numerator
of the time taken)

So, the efficiency of,

$$(B + C) = 60/12 = 5$$

$$C = 60/30 = 2$$

$$(A + B) = 60/(20/3) = 9$$

$$B = (5 - 2) = 3$$

$$A = (9 - 3) = 6$$

Statement I

The time taken by A to complete the whole work alone = $60/6 = 10$ days

So, statement I is correct.

Statement II

80% of the total work = 80% of 60 = 48

The time taken by B to complete 80% of the total work alone = $48/3 = 16$ days

So, statement II is correct.

Statement III

The combined efficiency of A, B and C together = $(6 + 3 + 2) = 11$

The time taken by A, B and C together to complete the whole work = $(60/11)$ days

So, statement III is correct.

Therefore, all statements I, II and III are correct.

Hence, the correct answer is option E.

16. Questions

Answer: B

Quantity I

The total investment made by P = $(X * 12) = \text{Rs. } 12X$

The total investment made by Q = $(X + 500) * (12 - 8) = (X + 500) * 4 = \text{Rs. } (4X + 2000)$

Now,

$$12X / (4X + 2000) = 9/4$$

$$48X = 36X + 18000$$

$$12X = 18000$$

$$X = 1500$$

$$\text{Quantity I} = (X/5) = (1500/5) = 300$$

Quantity II

The total investment made by M = $(1600 * 12) = \text{Rs. } 19200$

The total investment made by N = $(150\% \text{ of } 1600) * (12 - Y) = 2400 * (12 - Y) = \text{Rs. } (28800 - 2400Y)$

Now,

$$19200 / (28800 - 2400Y) = 8/5$$

$$96000 = 230400 - 19200Y$$

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$$19200Y = 134400$$

$$Y = 7$$

$$\text{Quantity II} = Y^3 = 7^3 = 343$$

Therefore, Quantity I < Quantity II.

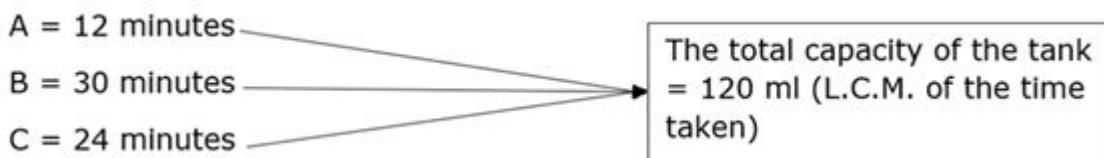
Hence, the correct answer is option B.

17. Questions

Answer: E

Quantity I

The time taken to fill/empty the whole tank by pipe,



So, the efficiency of pipe,

$$A = (120/12) = 10 \text{ (inlet)}$$

$$B = (120/30) = -4 \text{ (outlet)}$$

$$C = (120/24) = 5 \text{ (inlet)}$$

$$\text{The combined efficiency of pipe A and B together} = 10 + (-4) = (10 - 4) = 6$$

$$\text{So, the capacity of the tank filled by pipe A and B together} = 6 * (2n + 3) = (12n + 18)$$

$$\text{And, the capacity of the tank filled by pipe C alone} = (5 * n) = 5n$$

Now,

$$(12n + 18) + 5n = 120$$

$$17n = 102$$

$$n = 6$$

$$\text{Quantity I} = 10n = (10 * 6) = 60$$

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Quantity II

The total capacity of the tank = 60 ml

So, the efficiency of pipe X = (60/m)

And, the efficiency of pipe Y = 60/(m + 5)

The time duration for which pipe X remained open = (8 + 1) = 9 minutes

And, the time duration for which pipe Y remained open = 8 minutes

So, the capacity of the tank filled by pipe X = 9 * (60/m) = 540/m

And, the capacity of the tank filled by pipe Y = $8 * 60 / (m + 5) = 480 / (m + 5)$

Now,

$$540/m + 480/(m + 5) = 60$$

$$(540m + 2700) + 480m/(m^2 + 5m) = 60$$

$$1020m + 2700 = 60m^2 + 300m$$

$$60m^2 - 720m - 2700 = 0$$

$$m^2 - 12m - 45 = 0$$

$$m^2 - 15m + 3m - 45 = 0$$

$$m(m - 15) + 3(m - 15) = 0$$

$$(m + 3) + (m - 15) = 0$$

$$m = -3 \text{ or } +15$$

So, $m = 15$ (positive integer)

$$\text{Quantity II} = 4m = (4 * 15) = 60$$

Therefore, Quantity I = Quantity II.

Hence, the correct answer is option E.

18. Questions

Answer: A

Quantity I

Let the present age of husband and wife be $19m$ years and $16m$ years respectively.

So, the age of the husband at the time of the marriage = $19m - (3 + 2) = (19m - 5)$ years

And, the age of the wife at the time of the marriage = $16m - (3 + 2) = (16m - 5)$ years

Now,

$$(19m - 5) / (16m - 5) = 11/9$$

$$171m - 45 = 176m - 55$$

$$5m = 10$$

$$m = 2$$

So, the present age of the husband = $19m = (19 * 2) = 38$ years.

Quantity I = 38

Quantity II

Let the ratio of the present age of P and R be $3a$ years and $2a$ years respectively.

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Also, let the present of Q be b years.

Now,

$$3a - b = b - 2a$$

$$2b = 5a$$

$$a:b = 2:5$$

Let the values of a and b be $2c$ and $5c$ respectively.

So, the present age of,

$$P = 3a = (3 * 2c) = 6c$$

$$Q = b = 5c$$

$$R = 2a = (2 * 2c) = 4c$$

So, the ratio of the present ages of P , Q and $R = 6c : 5c : 4c = 6:5:4$.

Therefore, the required percentage = $(6 - 5)/5 * 100 = 1/5 * 100 = 20\%$

Quantity II = 20

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

19. Questions

Answer: B

Quantity I

Let the length of the platform be L meters

Now,

$$(400 + L)/120 = (300 + L)/100$$

$$40000 + 100L = 36000 + 120L$$

$$20L = 4000$$

$$L = 200$$

So, the length of the platform = $L = 200$ meters

Quantity I = 200

Quantity II

Let the speed of the train and the man be $14V$ km/hr and V km/hr.

So, the sum of the speed of the train and the man = $(14V + V) = 15V$ km/hr

Now,

$$225/15V = 3$$

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$$45V = 225$$

$$V = 5$$

So, the speed of the train = $14V = (14 * 5) = 70$ m/sec

$$\text{Quantity II} = (4 * 70) = 280$$

Therefore, Quantity I < Quantity II.

Hence, the correct answer is option B.

20. Questions

Answer: B

Quantity I

Let the total population of the village be 100K.

So, the number of illiterates = 25% of 100K = 25K

And, the number of literates = $(100K - 25K) = 75K$

The number of post-graduates = 25K

The number of graduates = $(75K - 25K) = 50K$

Now,

$$50K - 25K = 500$$

$$25K = 500$$

$$K = 20$$

So, the total population of the village = $100K = (100 * 20) = 2000$.

$$\text{Quantity I} = 2000$$

Quantity II

The ratio of the number of male and female employees = $100:50 = 2:1$. Let it be $2z$ and z respectively.

Now, the new number of male employees = $(2z - 500)$

And, the new number of female employees = $(z + 500)$

So,

$$(2z - 500) + (z + 500) = 3000$$

$$3z = 3000$$

$$z = 1000$$

So, the total number of employees in the company initially = $(2z + z) = 3z = (3 * 1000) = 3000$.

$$\text{Quantity II} = 3000$$

Therefore, Quantity I < Quantity II.

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Hence, the correct answer is option B.

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